BitLocker recovery key: how to get it safely

Article KB0010053 | Owner: Service Desk L1 | Subcategory: BitLocker recovery | Applies to: Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840, Windows 11 23H2 or 24H2, BitLocker-enabled devices managed by Microsoft Intune

Summary

This article explains how to retrieve a BitLocker recovery key safely when a Windows laptop prompts for the key at startup. It provides a self-help method for end-users who can verify their identity via the company self-service password reset portal, and a detailed procedure for Service Desk L1 agents to pull the key from Microsoft Intune and assist the user in recovering access.

Symptoms

- BitLocker recovery screen appears during boot
- Prompt: "Enter your recovery key to unlock this drive"
- 48-digit numerical key entry field
- No option to continue without the key
- Message may also show "Recovery key needed" after a BIOS/firmware update or TPM change

Common causes

- Forgotten BitLocker PIN or startup key
- BIOS/UEFI firmware update that changes PCR values
- TPM cleared or reset
- Major Windows update that alters boot measurements
- Hardware changes such as docking station or SSD replacement
- User entered incorrect PIN too many times causing lockout

Prerequisites

Before attempting recovery, ensure you have network connectivity and can authenticate to Acme Global resources. You will need your corporate credentials (username and password) and access to the Microsoft Authenticator app or an approved backup MFA method. If you cannot sign in, contact the Service Desk for assistance.

Self-help procedure - retrieve your own BitLocker key

If you are at the BitLocker recovery screen and remember your corporate credentials, you can retrieve the recovery key yourself using the Intune Company Portal website. This method works for any Acme-managed Windows laptop that is enrolled in Intune and has BitLocker enabled.

1. From another device (smartphone, tablet, or a colleague's PC) open a web browser and go to https://portal.manage.microsoft.com (the Company Portal).
2. Sign in with your Acme Global email address and password. Complete MFA using Microsoft Authenticator push or the backup SMS code.
3. Once logged in, locate your device in the "Devices" list. Click the device name to open its details page.
4. In the device details pane, find the "BitLocker recovery key" entry. Click "Show key" to reveal the 48-digit numeric key.
5. Copy the key exactly as shown (no spaces or extra characters).
6. Return to the locked laptop, enter the 48-digit key in the BitLocker recovery prompt, and press Enter.
7. Windows should unlock the drive and continue to the login screen. If prompted for a PIN, enter your usual BitLocker PIN.
8. After successful login, consider suspending BitLocker temporarily if you plan to perform a BIOS/firmware update, then resume it afterward (see the agent procedure for details).

Service-desk agent procedure - retrieve and provide the key

When a user contacts the Service Desk requesting a BitLocker recovery key, follow these steps to verify identity, retrieve the key from Intune, and guide the user to regain access. This process aligns with the standard L1 resolution: verify identity, provide key, confirm boot, and optionally suspend/resume BitLocker after a BIOS change.

1. Greet the user and confirm they are calling from an Acme Global device. Ask for their full name, employee ID, and the

laptop's asset tag or serial number.
2. Verify identity using one of the following methods: (a) Successful MFA-protected login to the Self-Service Password Reset portal (https://passwordreset.acme.example); (b) Confirmation of recent MFA push in Microsoft Authenticator; (c) Knowledge-based verification (e.g., date of birth, last 4 digits of employee ID) if MFA cannot be completed.
3. Open the Microsoft Endpoint Manager admin center (https://endpoint.microsoft.com) and navigate to Devices > All devices.
4. Locate the device by asset tag, serial number, or the user's display name. Click the device to open its profile.
5. Select "..." (More actions) -> "Export recovery keys" or locate the "BitLocker recovery key" field under Device configuration > BitLocker.
6. Click "Show key" to reveal the 48-digit recovery key. Copy the key to the clipboard.
7. Read the key back to the user slowly, confirming each group of six digits, or provide it via a secure chat (Teams) if the user prefers.
8. Instruct the user to enter the key at the BitLocker recovery screen and press Enter.
9. Once the drive unlocks and the Windows login appears, ask the user to sign in and verify they can access their desktop.
10. If the recovery was triggered by a BIOS/firmware update, advise the user to suspend BitLocker before future firmware changes: open an elevated PowerShell prompt and run `Suspend-BitLocker -MountPoint C: -RebootCount 0`. After the update, resume with `Resume-BitLocker -MountPoint C:`.
11. Close the ticket, noting the verification method used, the key retrieval source (Intune), and that the laptop booted successfully.

Verification

After providing the recovery key, the service desk must confirm that the user has regained access to the system and that BitLocker is functioning correctly. This step prevents repeated calls and ensures the encryption state is intact.

1. Ask the user to confirm they can log in to Windows with their usual credentials.
2. Have them open a Command Prompt (admin) and run `manage-bde -status C:` to verify that the drive shows "Protection On" and that the key protector matches the one just used.
3. If the user performed a suspend/resume action, re-run the status command to show that protection is back on.
4. Document the final status in the ticket and inform the user that their data remains encrypted and protected.

When to escalate

Escalate to the Identity & Access Management (IAM) team or Desktop Support if any of the following occur: the user cannot complete MFA verification, the Intune device record shows no BitLocker key, the recovery key fails to unlock the drive, or the BitLocker prompt persists after entering the correct key (indicating possible corruption or TPM failure). Include the asset tag, symptom description, steps already taken, and any error messages observed.

When to escalate

If the user cannot verify their identity via MFA or the self-service password reset portal, or if Intune does not contain a BitLocker recovery key for the device, escalate to the Identity & Access Management (IAM) team. Provide the user's name, employee ID, device asset tag, a description of the BitLocker prompt, and any error messages. If the key is present but fails to unlock the drive, or if the recovery screen reappears after correct entry, escalate to Desktop Support for possible TPM or disk corruption investigation.

Keywords: BitLocker, recovery key, Intune, Company Portal, BIOS update, TPM, encrypt, drive lock, 48-digit key, self-service password reset, MFA, Microsoft Authenticator, Endpoint Manager, manage-bde, Suspend-BitLocker, Resume-BitLocker